

Challenge (Habanero)

- Q1.** If two lines have the same slope $m = 2$, what can be said about them?
(A) They are perpendicular
(B) They are parallel
(C) They are intersecting
(D) They are skew
(E) They are coincident
- Q2.** A recipe for cake requires a ratio of 2 : 3 for flour to sugar. If 6 cups of flour are used, how many cups of sugar are needed?
(A) 6 cups
(B) 7 cups
(C) 8 cups
(D) 9 cups
(E) 10 cups
- Q3.** The equations $y = 3x + 2$ and $y = 3x - 4$ represent
(A) Perpendicular lines
(B) Parallel lines
(C) Intersecting lines
(D) Skew lines
(E) Coincident lines
- Q4.** An architect is designing a building with two parallel walls. If one wall is 15 meters long and the other is 20 meters long, what is the ratio of their lengths?
(A) 3 : 4
(B) 4 : 5
(C) 5 : 6
(D) 3 : 5
(E) 2 : 3
- Q5.** If $y = 2x + 1$ and $y = 2x - 3$ are parallel lines, what is the distance between them?
(A) 2 units
(B) 4 units
(C) $\frac{4\sqrt{5}}{5}$ units
(D) $\frac{4\sqrt{2}}{2}$ units
(E) 1 unit
- Q6.** The lines $y = x + 2$ and $y = x - 1$ are
(A) Perpendicular
(B) Parallel
(C) Intersecting
(D) Skew
(E) Coincident
- Q7.** A construction crew is building a road with two parallel lanes. If one lane is 500 meters long and the other is 600 meters long, what is the ratio of their lengths?
(A) 5 : 6
(B) 4 : 5
(C) 3 : 4
(D) 2 : 3
(E) 1 : 2
- Q8.** The equations $y = -2x + 1$ and $y = -2x - 3$ represent
(A) Perpendicular lines
(B) Parallel lines
(C) Intersecting lines
(D) Skew lines
(E) Coincident lines

Open Ended Questions (FRQ)

- Q9.** A chef is making a recipe that requires a ratio of 3 : 5 for ingredients A and B . If 12 cups of ingredient A are used, how many cups of ingredient B are needed? Show your work and explain your answer.
- Q10.** An architect is designing a building with two parallel walls. The equations of the lines representing the walls are $y = 4x + 2$ and $y = 4x - 1$. Find the distance between the walls and explain your method.

Chef's Tips

- Read each question carefully
- Use a ruler to draw diagrams
- Check your work before submitting